

An Investigation of Media Selection Among Directors and Managers: From "Self" to "Other" Orientation¹

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Abstract

The study investigates the media selection behavior of directors (executives) and managers through the use of multiple methods. The findings indicate the directors were more "self" oriented in their media choices, more often choosing media based on access/ease of use criteria, while the managers were more "other" oriented, more often making choices based on media richness/social presence criteria. These differences have implications in the interpretation of communication from directors and managers to the rest of the organization and suggest a model for understanding the use of "rich" and "lean" communication media. The literature review of the study

makes a major contribution by fitting together the multiple theories applied to the area and showing how conflicting results from all the established media selection theories make sense in different circumstances.

Keywords: Media selection, organizational roles, organizational communication

ISRL Categories: AF09, DC, DD02

Introduction

Communication activities account for a significant portion of the working time of managers. Previous research found that 75% of managers' time was spent in communication-related activities (Mintzberg 1973). A meta-analysis of media use studies was performed and found that managers spend 50% of the day communicating orally and 23% of the day communicating using text (Rice and Shook 1990). These findings are consistent with other studies of managerial use of time (Ives and Olson 1981; Kotter 1982; Poppel 1982). In fact, some authors have gone so far as to consider organizations as solely communication phenomena, that is, entities developed and maintained only through continuous communication activity among their participants (Farace et al. 1977; Weick 1979).

Managers dedicate a substantial amount of time to communication because they are making sense of their environments, coordinating and controlling internal activities, and making decisions (O'Reilly and Pondy 1979). While many activities are involved in communication, one that is of particular importance is media selection. High performing managers have been shown to match media with task (Daft et al. 1987), so individual performance is linked to judicious media choices. In addition, knowing where and how to obtain high quality information may allow managers to perform their jobs more effectively since evidence exists that the quality of information affects the quality of decisions (Porat and Haas 1969; Streufert

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1973) and having relevant information available favors the process of innovation (Zmud 1983). The importance of new media such as electronic mail are currently being explored. It is known that new media have the capability to improve and modify information gathering and dissemination strategies (Straub and Karahanna 1990) as well as to affect social interaction (Walther 1995) and to allow communication partners to express information that would not otherwise have been communicated (Sproull and Kiesler 1986).

Given the importance of both communication and media selection in organizations, it is not surprising that researchers have been increasingly interested in these phenomena. In spite of this interest, however, researchers have moved from simple cost/benefit analysis involving information access/quality tradeoffs to more complex social interaction theories over a period of 50 years without being able to substantially improve our ability to understand and predict media selection. This study investigates some reasons for the confusion in media selection, and suggests a methodology that is useful in such circumstances. It then applies this methodology through the analysis of empirical data. The paper proceeds as follows: first it explores the background of both the media selection literature and the relevant methodology literature. It lays out the project and the analysis of the data collected during the study. Then it looks at the literature again to help us make sense of existing theories in the light of the findings of the current study. Finally, the paper contains a discussion of the implications of the results of the analysis. The study indicates that directors (high in the hierarchy) use media differently than managers (lower in the hierarchy). It shows that directors think primarily about the effect of media choice on themselves while managers select media considering the effects on both themselves and others. Their selection criteria give rise to a model that explains some media selection behavior.

Media Selection Theories

Media selection theories are divided into two general categories based on the types of variables they include. The first category, referred to as trait theories of media selection, consists of two different explanatory approaches which consider media selection to be a function of traits of the media and characteristics of the task. The trait theories are discussed in detail below. The second category consists of approaches that delineate social influences affecting media use and show how users come to ascribe certain characteristics to the media and tasks that influence media selection. These approaches are referred to as social interaction theories. The interaction theories are examined after the trait theories in the following section.

Trait Theories of Media Selection

Two major trait theories have been applied to media selection. The first states that media selection is a function of a cost benefit analysis in which users try to attain an acceptable quality of information by exchanging information using media that require the least effort to access. This approach is referred to as the access/quality approach.

The second suggests that the exchange of information is only one task that occurs during communication. Other, equally important, aspects of communication involve matching the task with a medium based on the need to negotiate meaning or to feel that someone is physically present during the communication. Two theories have evolved in this category: media richness theory and social presence theory. First the access/quality theory, then the media richness and social presence theories are discussed below.

Access/quality

Media selection has been a subject of interest since the 1940s. The early studies in media selection were performed in order to understand sources of information. These studies were carried out for information warehousing agencies, such as libraries, that were trying to serve their patrons in the best manner possible. These early studies of the information gathering behavior of scientists and engineers were collected 20 years later (Menzel 1960). They concentrated on simple descriptive questions to help librarians and other providers of resources make available the most valued sources of information. During these studies it was discovered that, in addition to written sources, scientists and engineers rely heavily on informal face-to-face communication with colleagues both inside and outside the organization. In the late 1970s research showed that the accessibility of the source of information is more important than the quality of information gathered (Allen 1977). The predominance of accessibility over quality was later confirmed (O'Reilly 1982).

This line of inquiry was extended by information scientists who were interested in defining the use and characteristics of information. Information was considered to have objective quality characteristics such as relevancy (Zmud 1978), accuracy (Zmud et al. 1990), comprehensiveness and timeliness (Swanson 1987), which would affect the way information systems should be designed and implemented. Once installed, information systems were found to have accessibility characteristics that influence their use. Attributes such as convenience, easy access, and reliability were found to be important to users (Culnan 1984; Swanson 1987).

In this simple model of information gathering, accessibility is considered to be a cost, and quality is considered to be a benefit of information gathering, with the expected tradeoff in the direction of minimizing costs (Swanson 1987). In fact, the studies that rank variables by importance consistently indicate that accessibility is more important than quality in the choice of medium (Allen 1977; Culnan 1983;

Hardy 1982; O'Reilly 1982; Rice and Shook 1990). This finding implies that users will sometimes choose information of inferior quality simply because procuring it requires less work than procuring high quality information.

Media richness and social presence

The second set of trait theories involves two similar approaches developed independently by two groups, one in the U.S. and the other in Great Britain. An approach that came to be known as the media richness theory was developed by organization scientists (Daft and Lengel 1984, 1986). It started with information processing theory (Shannon and Weaver 1949), which states that the purpose of communication is to reduce uncertainty. It also includes the concept of equivocality reduction, which allows organizations to create a common understanding in ambiguous circumstances (Weick 1979). Reducing uncertainty and equivocality is necessary in order for the organization to accomplish internal tasks, to coordinate diverse activities and interpret the external environment.

Uncertainty is associated with lack of information, so to deal with uncertainty, organizations create structures such as formal information systems, task forces, and liaison roles that facilitate the flow of information. The role of media in uncertainty reduction is the ability to transmit the correct amount of information. Equivocality is associated with negotiating meanings for ambiguous situations. To deal with equivocality, the organization must find structures that enable rapid information cycles among managers so that meaning can emerge (Daft and Weick 1984; Weick 1979). The role of media in dealing with equivocality is the processing of rich information.² Media differ in their ability to convey rich information (Bodensteiner 1970). Media richness is the characteristic that describes this capacity for each medium with some media having

² Information richness is defined as the ability of information to change understanding within a certain time interval.

higher media richness than others based on the ability to give immediate feedback, the variety of communication cues, the personalization of the medium, and the language variety attainable.

In a parallel but separate effort, the Communication Studies Group in Great Britain identified the different types of communication activities associated with business meetings (negotiation, presentation of reports, delegation of work, information seeking, problem solving, forming impressions of others, giving information to keep someone in the picture) (Steinfeld 1985). Their objective was to determine whether some meetings might be more economically held using teleconferencing. The outcome was the theory of social presence (Short et al. 1976). Social presence is defined as the extent to which an individual psychologically perceives other people to be physically present when interacting with them. The theory says that people recognize that media differ in the amount of social presence they afford. They choose to use a medium based on the degree to which social presence is necessary for the particular communication task.

The media richness and social presence approaches are surprisingly similar, given that they resulted from two independent efforts: the media fall in the same order, whether they are arranged by richness or social presence (with face-to-face interaction at the top and written communication at the bottom) and many of the tasks talked about in the theories are similar. In addition, the approaches are intuitively appealing, since they reinforce our feeling that communication is not just about information but about the way information is communicated. However, when researchers began testing the theories empirically, while some evidence supported them, other results clearly indicated problems with the application of the theories.

Problems with media richness and social presence theories

As new media, particularly electronic media, became available in organizations,

researchers in the field of organizational communication and information systems found the explanations of task appropriateness and predictions of media selection based on the theories to be inadequate. The main type of counter evidence consisted of studies showing electronic mail was used for tasks that, according to the theories, would require media of markedly higher richness (such as face-to-face). One study reports a company in which managers are using e-mail, and sometimes using it as a rich medium (Markus 1994). Other research showed the same use of this media: messages of high socio-emotional content were being sent using electronic mail (a low social presence and low richness media) (Kiesler et al. 1984; Phillips 1983; Rice and Love 1987; Steinfeld 1985). Other evidence, such as reports of different uses of the same system in similar branches of the same organization (Rice and Manross 1987), or of different uses of the same system in different organizations (Zack and McKinney 1995) could not be explained by existing theories either.

The explanations found in the literature for these anomalous results are three-fold: (1) incorrect placement of e-mail on the media richness/social presence scale, (2) the idea of differences in the importance of some criteria in different situations, and (3) evolving understanding about how to use the media.

A first explanation for the apparent mismatches resides in the implicit comparison between the "richest," most capable of media (face-to-face) and all the other media. The media richness and social presence theories have been referred to as "cues filtered out" theories since they rank all media by the number of cues available in relation to face-to-face communication (Cuinan and Markus 1987). For example, the telephone allows verbal cues such as words, intonation, and silence, but cannot convey gestures or other visual cues. These comparisons may work well for traditional media, but electronic media have new capabilities (synchronous computer conferencing, ways of addressing communication, memory, storage and retrieval of communication, control over access to and participation in communication, lessening of status, and hierarchical cues) that

may effectively make them more capable of "rich" communication in users' perceptions. This explanation, then, states that these new capabilities were not taken into account when the criteria for classification on the media richness and social presence scales were established, resulting in incorrect positioning of new media on these scales.

A second reason for the conflicting results is based on the interplay between the task and the medium. Some evidence exists that participants will usually prefer a medium high on the richness scale when performing a task that they rank high in equivocality. Research has found that when equivocality is high, its effects supersede or mask the effects of other variables such as individual differences (Trevino et al. 1990b). This finding is supported by psychological research suggesting that individual differences are operative in determining behavior only when task demands and situational circumstances are weak (Mischel 1973; Mischel et al. 1973; Monson et al. 1982). However, participants who are choosing a medium for a task that is low in equivocality have a wider range of choices since, in effect, any medium is capable of communicating their message. Evidence exists that managers performing tasks that would normally require media of low richness (tasks that are low in equivocality) will often choose media that, on the contrary, are high in richness (Carlson 1993). They may be responding to variables such as job pressures (crises, unexpected problems, tight deadlines) and geographic dispersion (Steinfeld and Fulk 1986) or individual differences (Trevino et al. 1990b) that come into play in low equivocality situations, but that dictate preferences for media high in richness. This is true, even though rich media are considered more costly than "lean" media and are therefore deemed inefficient for tasks of low equivocality (Schmitz and Fulk 1991). This explanation implies that task descriptions are incomplete in current research, so that all influences on selection are not considered.

Third, the mismatch between the medium and the task may be explained by a difference in the perceptions of users about acceptable ways of using media. This difference in

perceptions could be explained by the time period required, after adoption of a technology, to develop a collective experience concerning media use. Some individuals perceive electronic media to be appropriate for a wider variety of communication activities as they gain experience with them (Hiltz and Turoff 1981; Rice 1984; Steinfield 1985). Subsequent research also found that the media richness theory applied to communication only in the early time periods (Rice et al. 1994). Research has shown that face-to-face communication is not more intimate and sociable than computer-mediated communication since participants found computer-mediated communication to be more dominant than face-to-face initially, but the effect dissipated over time (Walther 1995). One study (Carlson and Zmud 1994) indicated that the richness of a medium may depend less on the characteristics of the channel, and more on the perceptions of the user, experience and knowledge of the subject topic, and experience with the communication partner(s). This explanation implies that the difference in perceptions of users about acceptable ways of using media can also be explained by an emerging understanding of media use that goes beyond the usual technology adoption period (Orlikowski 1992; Poole and DeSanctis 1990). This area of investigation leads directly to the social interaction theories, which postulate a social influence on media selection and thus, answer the problems with media richness/social presence theories using another explanation.

Social Interaction Theories

The application of social interaction theories responded to the problems with media richness/social presence by postulating that media selection, like most tasks in organizations, is influenced by a combination of social forces. This means that, while the media richness/social presence scale matching tasks with media would apply in most cases, it is perfectly predictable that some groups or individuals will define either tasks or media traits differently, thus explaining the problems with

media richness/social presence theories. Social interaction theories were thus introduced in order to explain the major problems with media richness/social presence theories. As described in the following paragraphs, these theories have introduced their own set of problems.

The social interaction theories all approach the media selection area from a common point, but differ in their explanation of the mechanisms by which social influence happens. The fields of organizational communication and organizational behavior have engaged in alternate explanations for behavior influenced by subjectivity, socially constructed reality, and retrospective rationality (Fulk et al. 1987; Miller and Monge 1985; Putnam and Pacanowsky 1983; Zalesny and Farace 1986). Many of these theories, referred to as the social interaction theories, are based on symbolic interactionism.³ For organization studies, symbolic interactionism is based on the premise that organizations are webs of interaction, and the basis for interaction among members is a shared system of meaning. Each of the theories ascribes the creation of shared meaning among situated actors to social interaction, but their major contribution to the field of media selection is the description of different ways in which shared meaning is created. Three major theories have been applied to the field of media selection: structural symbolic interactionism, social information processing, and adaptive structuration theory.

Symbolic interactionism (Trevino et al. 1987) emphasizes social context as a source of influence in media selection. One framework based on symbolic interactionism is structural symbolic interactionism (Stryker and Statham 1985). According to structural symbolic interactionism, media selection behavior is determined at times by external factors such as situational constraints, and at times by social interaction factors. This approach incorporates

situational variables such as distance, time pressure, accessibility and connection of communication partners to the medium (Trevino et al. 1990a) as constraints.

Another theory based on symbolic interactionism is social information processing (Fulk et al. 1987). The basic premise of social information processing is that meaning is socially constructed (Pfeffer 1982; Salancik and Pfeffer 1977, 1978; Zalesny and Farace 1986). Objective characteristics and constraints in the work environment influence perceptions and behavior, but information provided by the social environment is of at least equal importance. This theory describes the specific mechanisms by which the social environment influences the formation of attitudes and behaviors through four types of communications from coworkers: (1) overt statements, (2) calling attention to certain aspects in the environment simply by communicating about them (which gives these aspects more saliency), (3) providing interpretations of events in the workplace, and (4) calling attention to particular deprivations and fulfillments in the environment. This means that coworkers, including superiors, contribute to perceptions by establishing criteria for media assessment, directing attention to salient media characteristics, and guiding interpretation of objective characteristics relative to those criteria.

While structural symbolic interactionism and social information processing imply a sequential relationship among the variables used in the selection process, structuration theory (Giddens 1979) describes social interaction as an iterative process. Structuration theory is the basis for the theory of adaptive structuration (Poole and DeSanctis 1990) in which group structure is both a medium and an outcome of group interaction with technology. According to this theory, all groups select specific features of technology to use in interaction and thus shape the effects of that technology on the group. The structuration model argues that technology is both a medium and an outcome of human action and that, similarly, properties of institutions are both an influence on and are influenced by human interaction with technology (Orlikowski 1992).

³ Symbolic interactionism is a theoretical framework that views society as a dynamic web of communication (Blumer 1969; Cooley 1902; Dewey 1922; Mead, 1934). Society is interaction, and interaction is symbolic since, through their interactions, people assign meaning to things.

In addition to these theories, other authors have proposed models that explain how social interaction works. A model based on the decision-making process proposes that established interaction patterns are influential in the selection process (Saunders and Jones 1990). It has been shown that media such as e-mail, previously thought to be appropriate for simple, impersonal communication, can be used for richer communication when the social processes surrounding media use define it as a rich medium (Markus 1994). The mechanism responsible for this phenomenon has been explained using hermeneutic analysis to discover that e-mail recipients are not passive, they actively assign meaning to each communication (Lee 1994).

The social interaction theories all propose mechanisms by which social influences act on users to guide their selection of media. This serves as an organizing influence for understanding media selection. However, they also acknowledge a set of variables that act as constraints on media decisions without proposing a hierarchy of choice. This means that the theories have served to introduce even more variables in media selection without improving our ability to predict selection behavior. In fact, early investigations into the characteristics of information, joined with the complexity of the media richness/social presence theories to include social interaction theories, made research very complex. Table 1 shows a sample of 22 variables that are considered to be important in media selection. These traits include task (Steinfeld 1985), job pressure (Steinfeld and Fulk 1986), geographic dispersion (Steinfeld and Fulk 1986; Trevino et al. 1987), job categories (Rice and Manross 1987; Rice and Shook 1990), complexity (Culnan 1983), ease of use (Culnan 1984), perceived attitudes and behavior of communication partners (Fulk et al. 1995), social influences (Schmitz and Fulk 1991), social context (Rice et al. 1994; Zack and McKenney 1995), job categories and organizational levels (Jones et al. 1988–89; Rice and Aydin 1991; Rice and Shook 1990; Zmud et al. 1990;) the symbolic cues provided by media (Trevino et al. 1987), and information relevancy (Zmud 1978).

Partially in response to the proliferation of variables, researchers have started trying to make sense of the conflicting results and multiple theoretical explanations by applying additional theories to the area. They have brought together different elements of influence and proposed ways the elements work together to determine media selection. A model of technology use has been proposed (Schmitz and Fulk 1991) that integrates aspects of media richness, social learning, social information processing, and symbolic interactionism. The model postulates that medium expertise and social influence along with media richness are important in the decision to use a particular medium. A set of propositions have been developed (Saunders and Jones 1990) that link media and source selection to the different phases of decision making. In this model, since information acquisition is subject to the dynamic aspects of decision making, the phase of decision making during which the information is gathered is important in the choice of media. Thus, not only have the number of important variables proliferated, but the range of theories applied to media selection has expanded.

Situations like the one in the area of media selection, where conflicting results and confusion exist, are ideal for the practice of certain types of research methodologies. Appropriate methodologies combine the advantages of multiple methods to test theories (Jick 1979; Lee 1991). For the reasons discussed below, the methodology used in the present study conforms to this model.

Methodological Issues

Much attention has been paid recently in the IS literature to different approaches to research (Lee 1991, Lee et al. 1997, Orlikowski and Baroudi 1991; Sabberwal and Robey 1995). These approaches to research have traditionally been associated with specific data collection and analysis techniques. While a full discussion of the advantages and disadvantages of these methods is not within the scope of the present paper, the following

Table 1. Empirical Studies Investigating Media Selection

Variable	Article
Individual differences	Trevino, Lengel, Bodensteiner, Gerloff, Muir (1990a)
Perceived attitudes and behavior of communication partners	Fulk, Schmitz, Ryu (1995)
Job pressure	Steinfeld and Fulk (1986)
Geographic dispersion	Trevino, Lengel, and Daft (1987) Steinfeld and Fulk (1986)
Media experience	Rice, Kraut, Cool, and Fish (1994)
Social influences	Schmitz and Fulk (1991) Dean and Brass (1985)
Job categories	Rice and Shook (1990) Rice and Manross (1987)
Organizational levels	Zmud, Lind, and Young (1990) Jones, Saunders, and McLeod (1988–89) Fulk, Schmitz, and Steinfeld (1990) Hinds and Kiesler (1995) Markus (1994)
Social context	Zack and McKenney (1995) Rice, Krant, Cool, and Fish (1994) Fulk, Schmitz, and Ryu (1995)
Task	Steinfeld (1985)
Socio-emotional content	Steinfeld (1985) Rice and Love (1987)
Information quality	Zmud (1978) Zmud, Lind, and Young (1990) O'Reilly (1982)
Access quality	Swanson (1987) O'Reilly (1982) Culnan (1983, 1984) Zmud, Lind, and Young (1990)
Media richness	Schmitz and Fulk (1991) Markus (1994) Rice (1992) Rice, Kraut, Cool, and Fish (1994)
Message equivocality	Daft, Lengel, and Trevino (1987)
Symbolic cues provided by media	Trevino, Lengel, and Daft (1987)
Environmental task complexity	Culnan (1983)
Ease of use	Culnan (1984) Adams, Nelson, and Todd (1992)
Time	Trevino, Lengel, and Daft (1987) Rice, Kraut, Cool, and Fish (1994) Walther (1995)
Relevancy	Zmud (1978)
Stage of decision making	Jones, Saunders, and McLeod (1988–89)
Feedback	Zmud, Lind, and Young (1990)

discussion outlines how different data collection methods have affected results in the area of media selection.

The predominant data collection method used to investigate the access/quality and the media richness theory is self report, usually in the form of a questionnaire. The differences in results obtained using observation versus self-report have been documented. In a meta-analysis of 40 media studies (Rice and Shook 1990), differences were noticed in media use based on the method of reporting the use: when participants were observed or asked to keep a diary, the total use of oral channels was significantly higher than when multiple methods or self report was used. Similarly, managers reported perceptions of media characteristics that fit reasonably well in the media richness theory, but their actual use of the media was different from predictions that could reasonably be made based on the reported perceptions (Markus 1994).

The predominant method of data collection in the development of the theory of social presence was laboratory experiments. In the area of communication studies, lab experiments are often designed so the participants in the experiments do not know each other before communicating via the new medium. They are also not given any choice about using the medium, they have no alternative ways to communicate, and they are often unfamiliar with the technology (Shulman and Steinman 1977). In addition, there are no differences among the participants in task or status. This means that the results from lab experiments are most likely indicative of organizations in which members have equal status and similar tasks, and are geographically remote from each other so they have never met, have few opportunities to meet, and few alternative communication media (Culnan and Markus 1987).

Because no single method of data collection and analysis is devoid of disadvantages, one solution to the limitations of using only one method is to combine methods. Recently, there has been a movement to combine the techniques from the different research approaches (Lee 1991, Robey 1996).

Combining methodologies has many advantages. It provides the opportunity, at multiple places in the research and analysis, to verify conclusions. Specifically, the researcher can cycle through the different stages in multiple iterations and verify one set of conclusions against the understandings in the other stages. It also provides a set of checks and balances so that the advantages of one data collection or analysis technique off-sets the disadvantages of another.

The Research Project

While the study was intended to test the two main sets of theories that have been applied to explain media selection (trait theories and social interaction theories), given the complexity of the area the study was designed to collect data about topics that then seemed extraneous to the study. It has been confirmed that researchers can increase their degrees of freedom and "obtain more information from a single study" by simultaneously considering alternative explanations in the design of studies (Markus 1994). Another major study (Kaplan and Duchon 1988) illustrates this principle as the original research design included both quantitative and qualitative methods of data collection and analysis, and it was apparent that the two approaches were essential to understanding differences in the acceptance and use of computer systems. In order to fulfill this objective in the current study, data about communication incidents observed were collected and participant input was gathered using semistructured interviews rather than questionnaires to gather the most data possible. In this way, the data set included more information than that explicitly necessary for the designed test so additional ad hoc tests could be carried out. The next section gives an overview and explanation of the research study.

Methodology

The study was conducted in a public agency in an upper Midwestern state. The agency consists of three departments that collect information regarding state educational policies and manage various aspects of educational lending and administration. The agency includes 70 people. The top 10 positions exercise supervisory power over the other members of the organization. They are the only members of the organization with supervisory or managerial duties. This entire population, consisting of the three people at the head of the organization (directors) and the seven managers reporting to the directors, were chosen to be participants in the study.

During the year preceding the first data collection, the researcher worked in an advisory capacity in the agency and secured the permission of the agency executive director to solicit participants from among the agency management staff. Participation was encouraged by assuring the potential subjects that their personal privacy would be strictly respected and that the results would be reported in aggregate form only.

The agency was an appropriate site for the study of media selection which includes new technology since, in addition to the traditional means of communication, organizational members had access to electronic messaging.

Data collection

The data collection methods consisted of observation of the participants and subsequent interviews about the events logged during the observation. The resulting data are of two types: observations of the participants during normal work days and texts of tape-recorded interviews with the participants.

The observation data were recorded as the participants went about their daily routines. The observation of participants consisted of minute by minute records of the ways participants gathered and disseminated information.

Each participant was observed for two complete eight-hour work days making a total of 20 days of observation. This data showed the media selected for communication and the communication partner with whom the communication "incident" was carried out. The observations were divided into communication "incidents" based on media use. Each time the participant employed a different medium, or when the communication partner changed, an incident was logged. A total of 514 communication incidents were cataloged during the observation phase of the research.

The participants were interviewed about the observation data. The interviews were held individually with each participant from one to two weeks after the observation of the participant with each interview lasting from one to two hours. Because it was impossible to discuss the totality of communication incidents during the time available for interviews, from 10 to 20 incidents were chosen from the observation data for discussion with each subject. The incidents were chosen for discussion based on representativeness of the total base of incidents including media used and communication partner. For each participant incidents were selected to include use of all media and a broad range of communication partners, while maintaining the proportions of usual media use and usual communication partners. The selected incidents were discussed in depth in semi-structured interviews with each participant (see Appendix A). The interviews were tape-recorded, and the texts of the interviews were transcribed and entered into a PC-based content analysis program.⁴ In all, the participants were asked to discuss 135 of the 514 communication incidents during the interviews.

Analysis

The analysis included two steps. First, insight into media selection, based on observations

* TALLY is a content analysis program used extensively to organize and code text data. It allows words and phrases in the text to be marked and sorted in different manners so that major themes become apparent.

during data collection, was surfaced and confirmed using statistical analysis. The observation data were used to develop an interpretation of how the participants themselves act out their media selection choices. A cluster analysis of the observation data confirmed the interpretation of the underlying pattern of media use.

In the second step, the findings were verified: the coded interview data were combined with the observation data in a manner recommended by Strauss (1987) to arrive at an understanding of media selection, then the data were subjected to statistical analysis to confirm the understanding. The data analysis is discussed in detail in the following paragraphs.

Analysis step one: surfacing and confirming insight into participant behavior

During the observation phase of the study, the participants were actively observed. During this time, it was noticed that the people at the top of the organization (directors) and those reporting to them (managers) go about the communication process in different manners. The hierarchical position of the participant seemed to influence communication behavior, although it was not possible at this point to precisely express this difference. It seemed that directors (the people highest in the organization) were somehow less "calculating" or "concerned" about their communication behavior. This point of view was confirmed by comments such as the following from a director (Director participant 334).

I write letters, publish reports, newsletters, I go to meetings, I give speeches orally, I phone people, so we take advantage of anything that will work, I'm a total pragmatist on that point.

In comparison, the remarks from managers tended to include such observations as the following (as the participant [338] adjusts the time of his coffee break to correspond to the time he thinks he may have the best chance of

catching someone in his office for an informal chat).

Okay, oftentimes I have minor items I would like to discuss with Larry⁵ . . . and since periodically I go over and get a cup of coffee, rather than do it earlier—I know . . . about the time Larry comes in, if he's coming in. So I will oftentimes delay that time. That way I can check to see if Larry is in . . . and then I can just discuss the topics which we should probably talk about.

If this insight proved to be true, it would lead the two groups to exhibit different communication patterns. Testing this insight involves using observational data. During the 20 days of observation, each participant was observed for two complete days and a total of 514 communication incidents that were initiated by the participants were recorded. These incidents, during which the participants selected media that they deemed appropriate for particular communication partners, were analyzed. The incidents were coded for type of media used and person with whom the communication was carried out (communication partner). The media categories were face-to-face, telephone, electronic mail, and written communication. Communication partners were categorized as superior, peer, subordinate, secretary, and outside organization. A cluster analysis was run on a table consisting of the percentage of use of the four media with each of five communication partners for each subject, making 20 data points per subject. The goal of cluster analysis is to identify homogeneous groups or clusters. In cluster analysis, membership in the groups is unknown at the beginning of the analysis and, during the analysis, the behavior of the subjects is compared and those with similar behavior are grouped together. The results are shown in Figure 1.

The cluster analysis in Figure 1 indicates that the participants form two distinct groups based on their media selection patterns. The largest group contains the seven participants who work at the level of management below the

⁵ All names have been changed to protect the identities of the participants and the people with whom they communicated during the study.

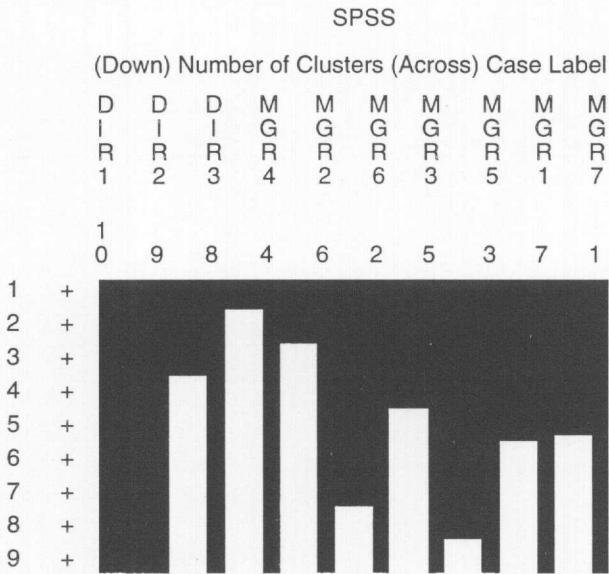


Figure 1. Cluster Analysis of Communication Behavior

The columns of the figure correspond to the communication behavior of the directors and managers. The first column is one of the directors (Director 1), the last column is one of the managers (Manager 7). The behavior of each subject is called a case. There are 10 cases since there were three directors and seven managers in the study. The figure is read from bottom to top. Each case is initially represented by a solid black line separated by white space. As the clusters are made and joined together, the graph becomes progressively more black (since spaces are no longer separating the solid lines). Each case is considered initially as an individual cluster. Because there are 10 managers, there are 10 clusters in row 10 (not shown by the package since that row gives no additional information). The plot starts at the bottom with row nine by showing the result of combining the two cases that are most similar into a cluster, making nine clusters. The bottom row of the figure shows these nine clusters, with Director 1 and Director 2 in the same cluster. At row eight in the graph, Manager 3 and Manager 5 are joined into the same cluster. The graph proceeds in this manner until, in row two, all three directors are in the same cluster, and all managers are in another. At this row the graph shows the communication behavior of the managers grouped into one cluster and the communication behavior of the directors grouped into another cluster.

directors. The smallest group contains the three participants who work at the director level.

The cluster analysis indicates that media selection patterns within each group are homogeneous while differences exist between the two groups. This test validates the understanding gained by the researcher during data collection. It shows that the media selection patterns of directors and managers are different. Directors behave in ways similar to each other, and managers behave similarly to each other. It does not show, however, the motivation for this behavior. In fact, while hierarchical level or status in the organization appears to be relat-

ed to communication behavior, other factors may also explain the media choices of participants as seen from the literature.

Analysis step two: explaining the differences in communication behavior

This analysis step was designed to explore all the reasons for media selection that differentiate directors from managers. The analysis described below incorporates the reasons directors and managers give for media selection.

The data to test this research question were collected during interviews with the participants. The interviews were based on semi-structured interview schedules (see Appendix A) that enabled the participants to explain the reasons they selected certain media in situations that had occurred during the two days of observation of each subject. A total of 135 incidents were discussed, with each participant discussing from 10 to 20 communication incidents. The participants were not prompted for answers, but gave spontaneous reasons for media selection in those incidents. The interviews were transcribed and content analyzed as described in the following paragraphs.

The reasons for media selection were coded into the descriptive categories shown in Appendix B. This coding was carried out according to the approach developed by Glaser and Strauss (Glaser and Strauss 1967; Strauss 1987). Using this method, the analysis starts with the data elements being assigned codes that are descriptive only. An examination of the codes in Appendix B shows the descriptive nature of this first coding step. The method, recommended by Strauss, gradually moves toward higher-level concepts, achieved by subsequent grouping of codes into higher-level categories.⁶

This method involves successive coding efforts starting with codes that are entirely descriptive and grouping the descriptive codes into categories that are progressively more abstract. The successive categories in this step grew out of a process referred to as "selective coding," which involves inspecting the existing codes and grouping them into higher level categories. This process, as shown in the following analysis, can involve multiple iterations as the higher level categories move farther from the literal meaning of the preliminary codes. In this process, the preliminary descriptive codes were assigned categories that reflected participant concern with

different aspects of the communication incidents. These categories evolved naturally out of the descriptive codes, and were mutually exclusive and all inclusive. The categories evolved as the preliminary descriptive codes fit into groupings:

1. Access/ease of use
2. Participant-related
3. Information-related
4. Situation-related
5. Media-related
6. Communication partner-related
7. Relationship/interpretation-related

All of the codes from the first descriptive step (found in Appendix B) are rearranged into the categories of the second level of coding in Appendix C. Since the observation data showed that status or hierarchical position in the organization is significant in influencing media selection, the code groupings are displayed in director and managerial categories in Appendix C. Appendix C is summarized in Table 2.

Based on the second-level categories, the most common reason for media selection is the search for the fastest, easiest way of communicating (38% and 30%). This category contains reasons that touch on convenience, cost, ease of use, and ease of access. It supports the *access vs. quality* trait theory of media selection and shows the great importance of the access trait and the reason that access was one of the first traits to be discovered as media selection was investigated.

The second largest category has to do with the relationship/interpretation-related reasons participants gave for using the media. This category (at the bottom of the table, representing 14% and 22% of the reasons) mainly contains entries that show an awareness of relationships with others and the need to sort out the meanings of situations with other people (reassuring, gaining consensus, personal connections, influencing, intimidating, disagreeing, dealing with complexity, observing non-verbals). The communication partner category (11% and 15%) contains reasons that have to do with the other person in the communication

⁶ For example, convenience was mentioned eight times, and it is simply coded as "convenience" in Appendix B. In the next level of analysis (see Appendix C), "convenience" is grouped with other descriptive codes in a higher-level category named "Access/Ease of Use."

Table 2. Second-Level Coding Categories

Reasons for Media Selection	Directors		Managers	
	No. Codes	%	No. Codes	%
Access/ease of use	32	38	85	30
Participant-related	12	15	19	7
Information-related	11	13	31	11
Situation-related	1	1	16	6
Media-related	7	8	29	10
Communication partner-related	9	11	43	15
Relationship/interpretation-related	12	14	64	22
Total	84	100%	287	100%

(considerations specific to the particular communication partner, opportunities that are created to interact with the communication partner, and specific statements that the preferences of the communication partner were important in the media selection decision). These two categories reflect the *media richness/social presence* theory of media selection since they involve tasks that are best thought to require rich media and to result in high levels of social presence. Together these two categories represent 25% of the reasons for directors and 37% for managers. The importance of these categories also shows one reason that the media richness and social presence trait theories were supported by empirical research.

Less prominent categories include those with reasons that are strictly related to characteristics of the media (asynchronous capabilities, paper trail, ability to show a document), those concerned mainly with characteristics of information (currency, type of information, informing another), and situation-related codes (job related, presence of researcher in situation, urgency). The final category includes codes that are related primarily to the participant (the media reflects the subject's philosophy of communication or other traits that are appropriate for the particular subject).

From the table, 38% of the codes represent the access/quality trait theory for directors and 30% for managers; 25% represent the media

richness/social presence trait theory for directors and 37% for managers; 37% represent "other" codes for directors and 33% for managers. It is possible that the existence of such a large number of codes that are not in the trait theories shows one reason that those theories did not consistently explain media selection behavior. It is also possible that having the remaining codes about evenly distributed between two of the trait theories explains why neither of these has been consistently the dominant theory.

Analysis of the codes reveals an interesting pattern. Figure 2 shows a graph of the data given in Table 2. The reasons given by directors and managers are grouped into the seven higher-level analysis categories on the graph, which is arranged so that categories for which directors have higher numbers are on the left side of the graph, while those for which managers have higher numbers are on the right side of the graph. The graph shows, as does the literature, that access/ease of use is important in media selection for both directors and managers, although it is more important for directors. It also shows that relationships are important for both directors and managers, but more important for managers. The graph helps organize the categories visually. Inspection of the graph in Figure 2 shows that directors tend to cite access/ease of use and participant (self-oriented) reasons for media selection in a greater proportion of entries, while managers

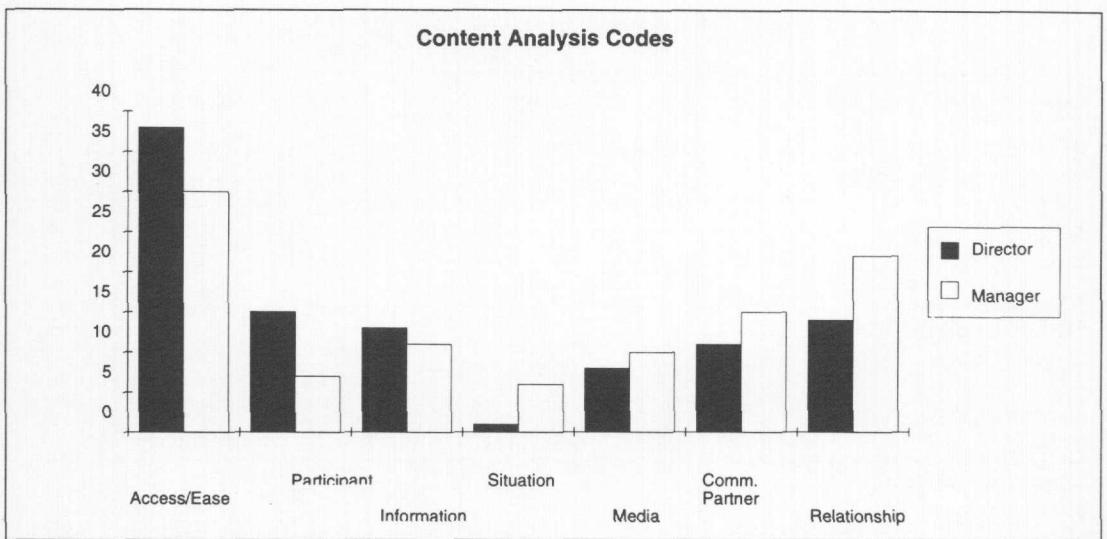


Figure 2. Graph of Second-Level Coding Categories

tend to cite relationship-related and communication partner (other-oriented) reasons more.⁷

Based on this observation, a final step of analysis grouped the seven categories into three, as shown in Table 3. The grouping is done along the lines of "self-oriented" versus "other-oriented" orientation. The "self-oriented" categories contain reasons that indicate the participants' own needs in communication (access, ease of use, subject's philosophy of communication, etc.). The "other-oriented" category includes consideration of the perceived preferences of the communication partner and relationship-oriented reasons. The remaining categories are grouped in the table into a "general" category. This category includes reasons specific to information, situation, and media.

A test of differences between proportions was used to statistically test the differences among the three groups of categories. This test is used to make inferences about data that is not continuous for which we wish to know whether

there are differences between the proportions of cases in categories (Chase 1984). The test is a one-tailed test, since the results would be expected to fall in only one direction. The alpha value was set at .05. The resulting z values are shown in Table 4.

The test results, shown in Table 4, indicate significant differences between the directors who chose media based on "self-oriented" reasons and managers who chose media for the same reason. The test also shows that a significant difference exists in the directors who choose media based on "other-oriented" reasons in relation to managers who choose media for the same reason. No significant difference was found in reasons based on general criteria. This supports the original feeling that the people highest in the hierarchy were less concerned or calculating about communication: they are less concerned with the effect of the communication on their communication partners. This was true even though both directors and managers were highly concerned about both the access and the relationship issues (see Figure 2). The text from the interviews supports this orientation as shown in the typical comments from directors below:

⁷ While some research shows that women tend to be relationship-oriented and men tend to be self-oriented, the composition of the sample makes this explanation invalid for the present study since all the directors and managers except one were male.

Table 3. Final Grouping of Coding Categories

Reasons for Media Selection	Director	Managers
	No. Codes	No. Codes
"Self-Oriented" Category		
Access/Ease	32	85
Subject	12	19
Category Total	44 (52.4%)	104 (36.2%)
"Other-Oriented" Category		
Communication Partner	9	43
Relationship	12	64
Category Total	21 (25%)	107 (37.3%)
"General" Category		
Information	11	31
Media	7	29
Situation	1	16
Category Total	19 (22.6%)	76 (26.5%)
Total	84	287

Table 4. Results of Difference of Proportion Test

Category	z value
Self-Oriented Category	2.627*
Other-Oriented Category	2.225*
General Category	.740

*Significant at .05 level.

Director participant 336: You can use voice mail from home, you can use voice mail from Washington, D.C. . . . you can use it on Sunday, anytime.

Director participant 334: The people that I'm physically near, because they're within 30 steps, I tend to take those steps. The communication is effective that way, it's fast, I don't bother writing notes, I just drop it on his desk. Speed and efficiency, that's all.

These types of remarks are contrasted with the remarks from managers.

Manager participant 331: Now Barbara, on the other hand, structurally reports to me. . . . She's very efficient, there's really no reason

for me to probably ever talk to her. I mean, she knows what the duties are and she knows how to do them. . . . But I think it's better to just have a little teeny contact with people every now and then, again sort of as more of a confirmation of the group as opposed to exchanging information.

Manager participant 337: I choose my weapon for the task at hand, and for that reason, I cannot say e-mail is always more efficient, or I use e-mail all the time on this, or I use this and this. If I use e-mail for Sally, I know that Sally sits there with one eye on the screen, and the moment something comes up, she cuts immediately out of what she does, she looks at the e-mail, and two minutes later she's at my door, and half an hour

is shot. So there, I don't use e-mail at all, I always see her face-to-face, and I go to her office. That way I control the length of the communication.

Re-examination of the literature: importance of the status variable

The importance of status, while mentioned by some authors, did not appear as a major theory of media selection. The literature that addresses status now takes on added importance and could help guide understanding the findings of the current study. Researchers who have included the hierarchical positions (status) of their participants as a variable have often found it to be significant (Zmud et al. 1990). It has been reported that 20% of the differences in use of electronic mail was due to hierarchical level (Fulk et al. 1990). In general, studies have found that when executives and managers are grouped together, they use high richness media more than those lower in the hierarchy (Rice and Shook 1990). The present study, however, distinguishes between those at the top of the organization and those who report to them. This is an important distinction, since the three people at director status were the highest in the organization, and the seven at manager level reported to them. Only two studies were found that distinguish between executives and managers. One studied managers in an energy company (Jones et al. 1988-89). The executives at the highest level include participants at the vice-president and executive vice-president level. The study also included participants at middle and lower levels in the organization. It found that upper-level managers (executives) use written media more than other managers, middle-level managers use phones and written media, and lower-level managers use face-to-face more. The use of asynchronous technology by those high in the hierarchy was confirmed in the second study (Hinds and Kiesler 1995), which found that there are differences in use by level of management. Upper-level administration use asynchronous technology more than those at lower levels (in this case the technology was voice mail).

The findings from these two studies do not support the media richness/social presence theory. That theory would predict that top-level managers would have more ambiguity and complexity in their environments so they would prefer richer face-to-face communication to leaner asynchronous communication. The social interaction theory would also predict more use of face-to-face. The hypothesis is that top management has access to any type of medium they choose, and since they use media to send symbolic signals in addition to information, they are very sensitive to the signals communicated within each medium (Trevino et al. 1987). Since their positions symbolically represent part of company culture, executives could be expected to use rich media more frequently.

These studies do, however, reflect the findings of the present study since participants lower in the hierarchy make more use of "other-oriented" or relationship building/interpretation developing criteria for media selection, while directors tend to choose media based on "self-oriented" reasons.

Discussion

The first section below discusses the contributions of the study to research in the area of media selection. The second section discusses the limitations of the study due to the design and some offsetting advantages. The final section suggests future research and implications for practice.

Contribution

This study consolidates the work of many researchers over the last 40 years by using a multiple methods design that allowed behavioral results to be explained using interview data. The study has made three major contributions to the media selection literature.

First, it has reviewed the previous work in media selection, organized, and explained it.

The literature review illustrates that the basis for the major trait theories of media selection (access/quality and media richness/social presence) are well founded. Accessibility includes concepts such as ease of use and convenience which are part of the "self-oriented" set of criteria in the present study. Media richness and social presence theories are built on using rich communication media to build relationships and interpret meaning. These concepts are found in the "other-oriented" set of criteria in the present study. In fact, studying the concepts underlying the "self-oriented" variables, it becomes clear that the emphasis is on ease of use and accommodation to the exigencies of the sender. When the underlying concepts of the "other-oriented" variables are examined, it becomes clear that the emphasis is on accommodating the communication partner. These notions are true to the concepts of accessibility and high media richness/social presence situations. While the directors and managers are different in their tendencies, as shown by the test of proportions, they both use predominantly accessibility and relationship criteria for decisions. This observation shows why the trait theories (access vs. quality and media richness/social presence) were the first to be discovered as researchers investigated the area of media selection and how significant results were obtained in both trait theories.

Second, the study has illustrated, through the large number of variables that are important in the media selection process, at least one of the reasons that the area of media selection is so complex. It has also shown why consistent explanations were not found for media selection using the trait theories, since about 25% of the reasons given for media selection are not in a major trait theory. The use of multiple methods produced results that would not have been obtained using a single method analysis.

Third, and perhaps most importantly, the study has created the basis for a model of media selection decision making. It shows that directors tend to select media based on accessibility more than managers, and managers tend to select media based on relationships and interpretation more than directors. In other words, directors think primarily about the effect

of media choice on themselves while managers select media considering effects on both themselves and others. Stated in another way, the communication needs of directors allow them to select the easiest way of communicating, while those of managers require them more often to substitute other selection criteria. This observation is confirmed by results from the literature showing the impact of organizational status on behavior, in which those at the top of the organization more often choose media considered to be low in media richness, and those at the managerial level are more likely to use media that allow rich communication.

These facts lead to a model for media selection supported by the literature. The model states that the most desirable characteristic for communication is media accessibility, but that this variable is over-ridden at times by the presence of certain other needs, such as the need to communicate rich information to build relationships and interpret situations. According to the model, while managers may prefer easily accessible media, they have different needs than directors, and they use richer media to achieve their goals. The model is useful because it suggests the order of importance of some of the variables in the decision process and also because it supports the findings in the literature. The model shows that many variables are important in the media selection process. It accommodates the trait theories of media selection because it suggests that easy accessibility is the default selection criterion with other variables such as relationship building being more important when they are present. It also accommodates the social interaction theories since it leaves open the question of social influence in the prioritization of selection criteria other than accessibility.

Limitations

The research design is not without disadvantages. The intrusive nature of the on-site data collection effort by an observer may have been stressful for participants. Because of this stress and because of a desire to please the

observer or perform socially desirable acts, it may be that the behavior of the participants differed during observation from the behavior they manifest during regular office interactions.

A second disadvantage of collecting data by observing participants is the high level of time investment required. This situation limits the number of participants in any one study. The seven people in the study included all those qualified for the study in the organization (since they were the entire management staff) and the statistical analysis was done on the communication incidents and the reasons that they gave for media selection. Even so, this number remains a small group upon which to base conclusions.

There were some advantages offsetting the limitations. It was possible, because of the live nature of the field data, to observe phenomena not available in paper incidents. For example, in some cases the medium of first choice to acquire or disseminate information was not the medium actually used since the receiver of the communication was not accessible using that medium (for example, the phone was not answered in someone's office). In these cases, the participant usually selected another medium. In a questionnaire scenario, the participant may mark the preferred medium and the additional variables that govern the change from that medium to the medium actually selected may be lost.

The main strengths of the research design are the real nature of the communication incidents and the variety of different types of data collected in the interviews about the communication incidents. The following section explains the implications of the study findings for research and practice.

Implications for research and practice

These findings have important consequences for researchers in the area of media selection, for information systems professionals as they install systems and train users, and for execu-

tives and other organizational employees as they make media choices. Based on evidence from the present study, researchers should concentrate on understanding the "big picture" as communicators see it. The study found that the trait theories (access/quality and media richness/social presence) tell which of the variables are important. The study also discovered that testing one of the existing theories may give significant results, but does not improve the ability to predict media selection behavior. The theories must be combined into a working model of the decision process. One such process, suggested by the study, considers access to be the most desirable trait of media, since it is most convenient for the communicator. This trait, however, is over-ridden by other characteristics that, when needed, are more important. One such trait suggested by the study is the need for relationship building. Testing a model in which traits are traded off against each other is an important next step in understanding media selection.

Another part of the puzzle concerns the way communicators perceive the characteristics of the media. Social interaction theories propose many ways for these perceptions to be formed and shared. Testing the ways communicators change their perceptions of the media is a fruitful area of research with important implications for practicing information systems professionals. The social influence theories have also suggested that the perceptions of media may not be uniform throughout the organization. If we assume that the trait theories tell us what is important, and the social interaction theories tell us how the traits come to be considered by organizational members, we should design research to discover how this information is shared and whether everyone in the organization has the same ideas about media (or if each individual, department, etc. has its own communication "culture").

The findings of this study are also significant for information systems professionals. The manner of presenting the advantages of technology such as electronic mail may need to be tailored to the hierarchical level of the trainee. Based on this study, when training or installing communication systems for directors and

executives, the speed, accuracy, and ease of use advantages may need to be emphasized and installed or made available. When training managers, the emphasis should be on the ways of incorporating social cues into the communication.

For managers, employees, and executives, this research also provides a way to interpret the media choices of those from whom one receives communications. Receiving a request from an executive through a memo should be interpreted in the same manner as receiving a personal visit from a manager. In many organizations, memos and letters tend to be relegated to secondary priority, probably because of their lack of relationship building capabilities. According to the present study, this is a mistake of interpretation on the part of employees. Asynchronous communication from directors and executives is intended to convey the same request for cooperation as more personal communications from those closer to one in the hierarchy. The research also gives information to executives about the way in which their communication is perceived by others. For executives sensitive to the subtleties of communication, it can act as a guide to changes in communication patterns that will foster perceptions of a more relationship and people oriented nature on the part of those with whom they communicate.

The study contributes to knowledge in four ways. First, it provides data from a field setting that demonstrates how multiple theories are operative in media selection. Second, it shows that complexity may explain some of the contradictory results obtained in media selection research to date and gives an example of a methodology that deals with complexity. Third, it confirms that the trait theories of media selection have indicated variables important in media selection. Fourth, it suggests a decision model and approaches for researchers and practitioners in information systems.

References

- Adams, D.A., Nelson, R.R., and Todd, P.A. "Perceived Usefulness, Ease of Use, and Usage of Information Technology: A Replication," *MIS Quarterly* (16:2), June 1992, pp. 227-247.
- Allen, T.J. *Managing the Flow of Technology: Technology Transfer and the Dissemination of Technological Information within the Research and Development Organization*, MIT Press, Cambridge, MA, 1977.
- Blumer, H. *Symbolic Interactionism: Perspective and Method*, Prentice Hall, Englewood Cliffs, NJ, 1969.
- Bodensteiner, W.D. "Information Channel Utilization Under Varying Research and Development Project Conditions: An Aspect of Inter-Organizational Communication Channel Usages," unpublished doctoral dissertation, University of Texas at Austin, 1970.
- Carlson, J.R. and Zmud, R.L. "Channel Expansion Theory: A Dynamic View of Media and Information Richness Perceptions," *Academy of Management Best Papers Proceedings*, 1994, Dallas, TX, Aug 14-17, pp. 280-284.
- Carlson, P. "Media Selection for Information Acquisition and Dissemination: A Study of Managerial Preference," unpublished doctoral dissertation, University of Minnesota, 1993.
- Chase, C.I. *Elementary Statistical Procedures* (3rd ed.), McGraw-Hill Book Company, New York, 1984.
- Cooley, C.H. *Human Nature and the Social Order*, Charles Scribner's Sons, New York, 1902.
- Culnan, M.J. "Environmental Scanning: The Effects of Task Complexity and Source Accessibility on Information Gathering Behavior," *Decision Sciences* (14), 1983, pp. 194-206.
- Culnan, M.J. "The Dimensions of Accessibility to Online Information: Implications for Implementing Office Information Systems," *ACM Transactions on Office Information Systems* (2:2), 1984, pp. 141-150.
- Culnan, M.J., and Markus, M.L. "Information Technologies," *Handbook of Organizational*

- Communication: An Interdisciplinary Perspective*, F. Jablin, L. Putnam, K. Roberts, and L. Porter, (eds.), Sage Publications, Newbury Park, CA, 1987, pp. 420–443.
- Daft, R.L., and Lengel, R.H. "Information Richness: A New Approach to Managerial Behavior and Organization Design," in *Research in Organizational Behavior* (6), B. Staw and L.L. Cummings (eds.), 1984, pp. 191–233.
- Daft, R.L., and Lengel, R.H. "Organizational Information Requirements, Media Richness and Structural Design," *Management Science* (32:5), May 1986, pp. 554–571.
- Daft, R.L., and Weick, K. "Toward a Model of Organizations as Interpretations Systems," *Academy of Management Review* (9), 1984, pp. 284–295.
- Daft, R.L., Lengel, R.H., and Trevino, L.K. "Message Equivocality, Media Selection, and Manager Performance: Implications for Information Systems," *MIS Quarterly* (11:3), September 1987, pp. 355–366.
- Dean, J.W. Jr., and Brass, D.J. "Social Interaction and the Perception of Job Characteristics in an Organization," *Human Relations* (38), 1985, pp. 571–582.
- Dewey, J. *Human Nature and Conduct*, Holt, New York, 1922.
- Farace, R., Monge, P.R., and Russell, H.M. *Communicating and Organizing*, Addison-Wesley, Reading, MA, 1977.
- Fulk, J., Schmitz, J., and Ryu, D. "Cognitive Elements in the Social Construction of Communication Technology," *Management Communication Quarterly* (8:3), February 1995, pp. 259–288.
- Fulk, J., Schmitz, J., and Steinfield, C.W. "A Social Influence Model of Technology Use," in *Organizations and Communication Technology*, J. Fulk and C. Steinfield (eds.), Sage Publications, Newbury Park, CA, 1990, pp. 117–140.
- Fulk, J., Steinfield, C.W., Schmitz, J., and Power, J.G. "A Social Information Processing Model of Media Use in Organizations," *Communication Research* (14:5), October 1987, pp. 529–552.
- Giddens, A. *Central Problems in Social Theory*, Berkeley, University of California Press, Berkeley, CA, 1979.
- Glaser, B.G., and Strauss, A.L. *The Discovery of Grounded Theory: Strategies for Qualitative Research*, Aldine Publishing, Chicago, IL, 1967.
- Hardy, A. "The Selection of Channels When Seeking Information: Cost/Benefit vs. Least-Effort," *Information Processing and Management* (18:6), 1982, pp. 289–293.
- Hiltz, S.R., and Turroff, M. "The Evolution of User Behavior in a Computerized Conferencing System," *Communications of the ACM* (24:11), 1981, pp. 739–751.
- Hinds, P., and Kiesler, S. "Communication across Boundaries: Work, Structure, and Use of Communication Technologies in a Large Organization," *Organization Science* (6:4), July–August 1995, pp. 373–393.
- Ives, B., and Olson, M.H. "Manager or Technician? The Nature of the Information Systems Manager's Job," *MIS Quarterly* (5:4), December 1981, pp. 49–63.
- Jick, T.J. "Mixing Qualitative and Quantitative Methods: Triangulation in Action," in *Qualitative Methodology*, J. Van Maanen (ed.), Sage Publications, Beverly Hills, CA, 1979, pp. 135–148.
- Jones, J.W., Saunders, C., and McLeod, R. Jr. "Information Media and Source Patterns Across Management Levels: A Pilot Study," *Journal of Management Information Systems* (5:3), Winter 1988–89, pp. 71–84.
- Kaplan, B., and Duchon, D. "Combining Quantitative and Qualitative Methods in Information Systems Research," *MIS Quarterly* (12:4), 1988, pp. 571–586.
- Kiesler, S., Siegel, J., and McGuire, T. "Social Psychological Aspects of Computer-mediated Communication," *American Psychologist* (39), 1984, pp. 1123–1134.
- Kotter, J.P. "What Effective Managers Really Do," *Harvard Business Review* (60:6), 1982, pp. 156–167.
- Lee, A.S. "Integrating Positivist and Interpretive Approaches to Organizational Research," *Organization Science* (2:4), November 1991, pp. 342–365.
- Lee, A.S. "Electronic Mail as a Medium for Rich Communication: An Empirical Investigation Using Hermeneutic Interpretation," *MIS Quarterly* (18:2), 1994, pp. 143–157.

- Lee, A., Liebenau, J., and DeGross, J. I. (eds.) *"Information Systems and Qualitative Research,"* Chapman & Hall, London, 1997.
- Markus, M.L. "Electronic Mail as the Medium of Managerial Choice," *Organization Science* (5:4), 1994, pp. 502-527.
- Mead, G.H. *Mind, Self, and Society*, C. Morris (ed.), University of Chicago Press, Chicago, IL, 1934.
- Menzel, H. *Review of Studies in the Flow of Information Among Scientists*, Bureau of Applied Research, Columbia University, New York, 1960.
- Miller, K.I., and Monge, P.R. "Social Information and Anxiety about Organizational Change," *Human Communication Research* (11), 1985, pp. 365-386.
- Mintzberg, H. *The Nature of Managerial Work*, Harper and Row, New York, 1973.
- Mischel, W. "Toward a Cognitive Social Learning Reconceptualization of Personality," *Psychological Review* (80), 1973, pp. 252-283.
- Mischel, W., Ebbesen, E.B., and Zeis, A.R. "Selective Attention to the Self: Situational and Dispositional Determinants," *Journal of Personality and Social Psychology* (27), 1973, pp. 129-142.
- Monson, T.C., Hesley, J.W., and Chernick, L. "Specifying When Personality Traits Can and Cannot Predict: An Alternative to Abandoning the Attempt to Predict Single-Act Criteria," *Journal of Personality and Social Psychology* (13), 1982, pp. 89-111.
- O'Reilly, C.A. "Variations in Decision Makers' Use of Information Sources: The Impact of Quality and Accessibility of Information," *Academy of Management Journal* (25:4), 1982, pp. 756-771.
- O'Reilly, C.A., and Pondy, L.R. "Organizational Communication," in *Organizational Behavior*, S. Kerr, (ed.), Grid, Columbus, OH, 1979, pp. 119-150.
- Orlikowski, W.J. "The Duality of Technology: Rethinking the Concept of Technology in Organizations," *Organization Science* (3), 1992, pp. 398-427.
- Orlikowski, W.J., and Baroudi, J.J. "Studying Information Technology in Organizations: Research Approaches and Assumptions," *Information Systems Research* (2:1), March 1991, pp. 1-28.
- Pfeffer, J. *Organizations and Organization Theory*, Pitman Publishing, Boston, MA, 1982.
- Phillips, A. "Computer Conferences: Success or Failure?", in *Communication Yearbook 7*, R. Bostrom (ed.), Sage Publications, Newbury Park, CA, 1983, pp. 837-856.
- Poole, M.S., and DeSanctis, G. "Understanding the Use of Group Decision Support Systems: The Theory of Adaptive Structuration," in *Organization and Communication Technology*, J. Fulk and C. Steinfield (eds.), Sage, Newbury Park, CA, 1990, pp. 173-191.
- Poppel, H.L. "Who Needs the Office of the Future?", *Harvard Business Review* (60:6), 1982, pp. 146-155.
- Porat, A., and Haas, J. "Information Effects on Decision Making," *Behavioral Science* (14), 1969, pp. 98-104.
- Putnam, L., and Pacanowsky, M.E. (eds.). *Communication and Organizations: An Interpretive Approach*, Sage Publications, Beverly Hills, CA, 1983.
- Rice, R.E. "Evaluating New Media Systems," in *Evaluating the New Information Technologies: New Directions for Program Evaluation*, J. Johnstone (ed.), Jossey-Bass, San Francisco, CA, 1984, pp. 53-71.
- Rice, R.E. "Task Analyzability, Use of New Media and Effectiveness: A Multi-site Exploration of Media Richness," *Organization Science* (3:4), November 1992, pp. 475-500.
- Rice, R.E., and Aydin, C. "Attitudes Toward New Organizational Technology: Network Proximity as a Mechanism for Social Information Processing," *Administrative Science Quarterly* (36), June 1991, pp. 219-244.
- Rice, R.E., and Love, G. "Electronic Emotion: Socio-emotional Content in a Computer-Mediated Communication Network," *Communication Research* (14), 1987, pp. 85-108.
- Rice, R.E., and Mantross, G. "The Case of the Intelligent Telephone: The Relationship of Job Category to the Adoption of Organizational Communication Technology," in *Communication Yearbook 10*,

- M.L. McLaughlin (ed.), Sage Publications, Newbury Park, CA, 1987, pp. 727-742.
- Rice, R.E., and Shook, D.E. "Relationships of Job Categories and Organizational Levels to Use of Communication Channels, Including Electronic Mail: A Meta-analysis and Extension," *Journal of Management Studies* (27:2), March 1990, pp. 195-229.
- Rice, R.E., Kraut, R.E., Cool, C., and Fish, R.S. "Individual Structural and Social Influences on Use of a New Communication Medium," *Academy of Management Best Papers Proceedings*, 1994, pp. 285-289.
- Robey, D. "Research Commentary: Diversity in Information Systems Research: Threat, Promise, and Responsibility," *Information Systems Research* (7:4), December 1996, pp. 400-408.
- Sabberwal, R., and Robey, D. "Reconciling Variance and Process Strategies for Studying Information System Development," *Information Systems Research* (6:4), December 1995, pp. 303-327.
- Salancik, G.R., and Pfeffer, J. "An Examination of Need Satisfaction Models of Job Attitudes," *Administrative Science Quarterly* (22), 1977, pp. 427-456.
- Salancik, G.R., and Pfeffer, J. "A Social Information Approach to Job Attitudes and Task Design," *Administrative Science Quarterly* (23), 1978, pp. 224-252.
- Saunders, C., and Jones, J.W. "Temporal Sequences in Information Acquisition for Decision Making: A Focus on Source and Medium," *Academy of Management Review* (15:1), 1990, pp. 29-46.
- Schmitz, J., and Fulk, J. "Organizational Colleagues, Media Richness, and Electronic Mail: A Test of the Social Influence Model of Technology Use," *Communication Research* (18:4), August 1991, pp. 487-523.
- Schulman, A.D., and Steinman, J.I. "Interpersonal Teleconferencing in an Organizational Context," in *Evaluating New Telecommunications Services*, M.C. Elton, W.A. Lucas, and D.W. Conrath (eds.), Plenum Press, New York, 1977, pp. 399-424.
- Shannon, C.E., and Weaver, W. *The Mathematical Model of Communication*, University of Illinois Press, Urbana, IL, 1949.
- Short, J., Williams, E., and Christie, B. *The Social Psychology of Telecommunications*, John Wiley and Sons Ltd., Chichester, England, 1976.
- Sproull, L., and Kiesler, S. "Reducing Social Context Cues: Electronic Mail in Organizational Communication," *Management Science* (32:11), November 1986, pp. 1492-1512.
- Steinfeld, C. "Dimensions of Electronic Mail Use in an Organizational Setting," *Proceedings of the Academy of Management*, 1985, pp. 239-243.
- Steinfeld, C.W., and Fulk, J. "Task Demands and Manager's Use of Communication Media: An Information Processing View," paper presented at the meeting of the Academy of Management, Chicago, 1986.
- Straub, D.W., and Karahanna, E. "The Changing Role of Telecommunications Technologies in the Workplace: E-Mail, Voice-Mail and FAX," Working Paper WP-90-10, Management Information Systems Research Center, University of Minnesota, Minneapolis, MN, 1990.
- Strauss, A.L. *Qualitative Analysis for Social Scientists*, Cambridge University Press, Cambridge, MA, 1987.
- Streufert, S.C. "Effects of Information Relevance on Decision Making in Complex Environments," *Memory and Cognition* (1), 1973, pp. 224-228.
- Stryker, S., and Statham, A. "Symbolic Interaction and Role Theory," in *New Handbook of Social Psychology*, G. Lindsay and E. Aronson (eds.) Random House, New York, 1985.
- Swanson, E. B. "Information Channel Disposition and Use," *Decision Sciences* (18), 1987, pp. 131-145.
- Trevino, L.K., Lengel, R., Bodensteiner, W.D., Gerloff, E.A., and Muir, N.K. "The Richness Imperative and Cognitive Style," *Management Communication Quarterly* (4:2), November 1990a, pp. 176-197.
- Trevino, L.K., Daft, R.L., and Lengel, R.H. "Understanding Managers' Media Choices: A Symbolic Interactionist Perspective," in

- Organizations and Communication Technology*, J. Fulk and C. Steinfield (eds.), Sage Publications, Newbury Park, CA, 1990b, pp. 71-94.
- Trevino, L.K., Lengel, R., and Daft, R.L. "Media Symbolism, Media Richness, and Media Choice in Organizations: A Symbolic Interactionist Perspective," *Communication Research* (14:5), 1987, pp. 553-574.
- Walther, J.B. "Relational Impacts of Computer-mediated Communication: Experimental Observations over Time," *Organization Science*, (6:2), March-April 1995, pp. 196-203.
- Weick, K. *The Social Psychology of Organizing*, Addison-Wesley, Reading, MA, 1979.
- Zack, M.H., and McKenney, J.L. "Social Context and Interaction in Ongoing Computer-supported Management Groups," *Organization Science* (6:4), July-August 1995, pp. 394-422.
- Zalesny, M.D., and Farace, R.V. "A Field Study of Social Information Processing: Mean Differences and Variance Differences," *Human Communication Research* (13), 1986, pp. 268-290.
- Zmud, R.W. "The Effectiveness of External Information Channels in Facilitating Innovation Within Software Development Groups," *MIS Quarterly* (7:2), June 1983, pp. 43-58.
- Zmud, R.W., Lind, M.R., and Young, F.W. "An Attribute Space for Organizational Communication Channels," *Information Systems Research* (1:4), 1990, pp. 440-457.

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Appendix A

Interview Schedule

Part I

How long have you worked for this company? _____

How long have you held your present position? _____

How many people report directly to you? _____

My study has to do with the ways people communicate. I am interested in face-to-face, phone, e-mail, letters, etc. In terms of these media:

What was typical about the days I observed you?

What was unusual or atypical about the days I observed you?

How would you define quality? Accessibility?

Part II

For each incident:

1. What is the objective of this incident?
2. How well do you know the other person?
3. Why did you use _____?
4. Can you think of any other way you could have gotten that information/gotten that message out?
5. Can you think of any way or place you could find better information?
6. What was the deciding factor in your choice of _____?
7. Can you think of any other reasons you used _____?

Appendix B

List of Descriptive Codes

Content Analysis Codes	
Code Meaning	No.
Access	21
Asynchronous medium	9
Build team	3
Build consensus	7
Clarify	8
Complexity	9
Convenience	8
Cost	3
Current	1
Disagree	3
Discuss	9
Ease of use	15
Efficiency	15
Expedite	12
Fast	43
Gather information	16
Influence others	6
Inform other	6
Intimidate others	2
Related to job	8
Location of subject	10
Convey negative message	3
Observe non-verbals	2
Take advantage opportunity	20
Paper document	21
Personal connection	7
Fits participant philosophy	2
Reassure	12
Respect communication partner wishes	24
Show document	6
Presence of researcher	8
Strong media for persuasion	3
Traits of subject	19
Type of information	19
Urgent situation	1
Warm media	2
Who is communication partner	8

Appendix C

Codes Grouped in Core Categories

Content Analysis Codes		
Code Meaning	Directors	Managers
Acces/Ease of Use Codes		
Access	6	15
Convenience	3	5
Cost	0	3
Ease of use	3	12
Efficiency	3	12
Expedite	1	11
Fast	16	27
Subtotal	32	85
Percent	38%	30%
Media-Related Codes		
Asynchronous	1	8
Paper	1	20
Show document	5	1
Subtotal	7	29
Percent	8%	10%
Information-Related Codes		
Current	1	0
Gather information	4	12
Inform other	2	4
Type of information	4	15
Subtotal	11	31
Percent	13%	11%
Situation-Related Codes		
Related to job	0	8
Presence of researcher	1	7
Urgent situation	0	1
Subtotal	1	16
Percent	1%	6%
Participant-Related Codes		
Fits participant philosophy	2	0
Traits of subject	6	13
Location of subject	4	6
Subtotal	12	19
Percent	15%	5%

Appendix C. Continued

Code Meaning	Directors	Managers
Communication Partner-Related Codes		
Who communication partner is	1	7
Take advantage opportunity	4	16
Respect communication partner wishes	4	20
Subtotal	9	43
Percent	11%	15%
Relationship/Interpretation-Related Codes		
Reassure	0	12
Build consensus	2	5
Personal connection	2	5
Complexity	0	9
Influence others	6	0
Intimidate others	0	2
Disagree	0	3
Clarify	0	8
Discuss	1	8
Strong media for persuasion	0	3
Warm media	0	2
Observe non-verbals	0	2
Convey negative message	0	3
Build team	1	2
Subtotal	12	64
Percent	14%	22%